

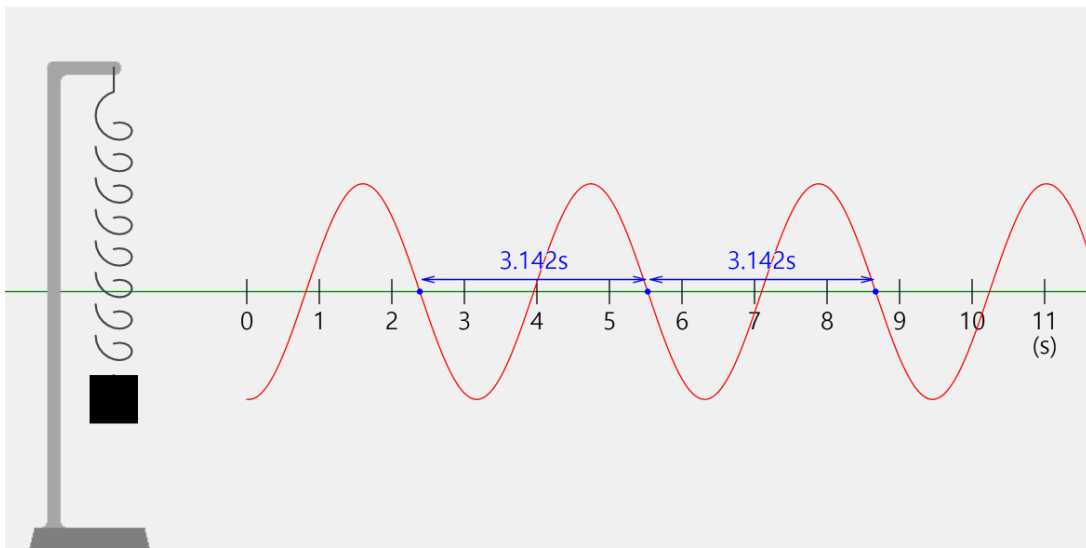
Graphical Analysis of a Spring Oscillator

ENGS121 Mechanics Lab Section D

17 April, 2025

1. Introduction

The experiment is done to observe the harmonic oscillations of a mass hang from a spring and displaced from the equilibrium position. In theory brought in school physics the mass of the spring is omitted. The formula of $T = 2\pi\sqrt{\frac{m}{k}}$ is used. However, the mass of the spring has its influence on the experiment outcomes. More accurate formula for the period is known as $T = 2\pi\sqrt{\frac{m+\beta\cdot m_{spring}}{k}}$. The experiment is conducted to check the formula and measure the coefficient β and the spring elasticity constant k .

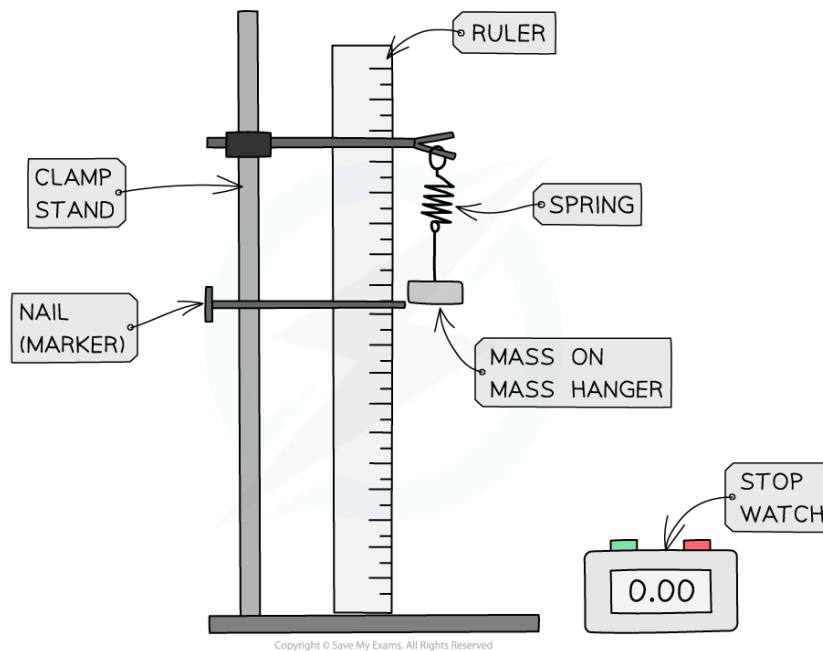


2. Measurements

Instruments

Name	Quantity	Properties
Stand	1 pc	Stand
Spring	1 pc	Lab Spring, with constant k
Weights	Set of Lab Weights	Different weights
Scales	1 pc	With an accuracy of 1 g
Stopwatch	1pc	Smartphone was used, accuracy of 0.01 s

Experimental Procedure



A weighted spring was mounted on a stand. Different measured weights were hung on the spring. As it was displaced from the equilibrium point, the time for 10 oscillations were measured with a stopwatch.

Mass of the Spring	27 g
Chosen Number of Spring Oscillations	10

Errors

Source of error	Type	Solution
Oscillations	Random	With lower masses, the spring was moving chaotically, more like swinging like a pendulum. The spring was released as vertically as possible to avoid swinging.
Reaction time	Random	Stopwatch was started only after the spring had completed one full oscillation to ensure the accuracy in measurements.
Weighting Scale Inaccuracy	Systematic	Can be solved by using another weighting scale.
Weights factory imperfections	Systematic	Can be resolved by replacing the weights.

Variables

Variable	Value	Resolution
Attached Weight	Controlled	1 g
Time of Oscillation	Measured	0.01 s
Number of Oscillations	Controlled	1

Spring Data Raw

m (g)	157	200	357	500	657	700	857	900	1000
t (s)	4.94	5.6	7.52	9.19	10.32	10.68	11.71	12.05	12.37

*Note that this is not the full table, masses below 150 g and higher than 1000g were used, however masses lower than 150 g were causing destabilization and chaotic oscillations, and masses higher than 1000 g would deform the spring and would change the spring elasticity which would have influence on our calculations. Hence the optimal range of the measurements were used for the table.

3. Data Analysis and Plots

In order to use graphical analysis, the equation was linearized by squaring both sides of the

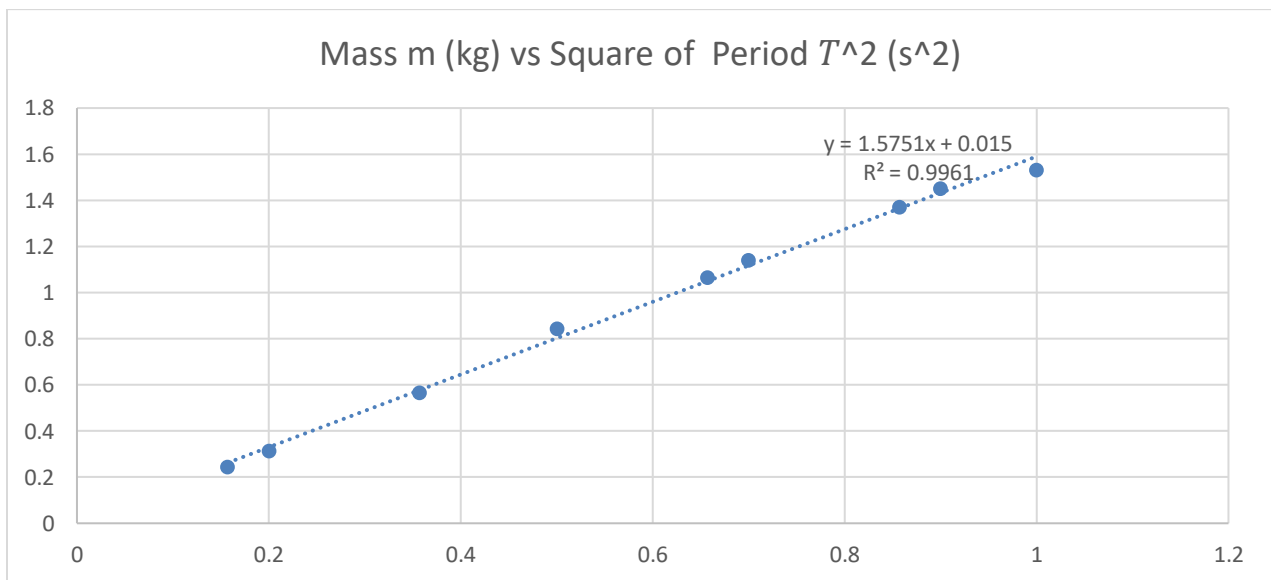
equation. The equation of $T^2 = \frac{4\pi^2}{k}m + \frac{4\pi^2 \cdot \beta \cdot m_{spring}}{k}$, was obtained, which follows the

$y = mx + b$ form. On the other hand, the equation of $T = \frac{t}{N}$, can be used to calculate the period.

Spring Data SI Units Converted, Period of Oscillations Calculated

m (kg)	0.157	0.2	0.357	0.5	0.657	0.7	0.857	0.9	1
T ² (s ²)	0.2440	0.3136	0.5655	0.8446	1.0650	1.1406	1.3712	1.4520	1.5302

Plot of attached weight and squares of periods



Data for Error Calculation

m ² (kg)	0.024649	0.04	0.127449	0.25	0.431649	0.49	0.734449	0.81	1
T ² (s ²) (pred.)	0.2623	0.3300	0.5773	0.8026	1.0498	1.1176	1.3649	1.4326	1.5901

The error in slope is

$$(\Delta k)^2 \approx \frac{1}{n \cdot (\overline{x^2} - (\bar{x})^2)} \cdot \frac{\Sigma(y_{predicted} - y_{experiment})^2}{n - 2}$$

The error in intercept is

$$(\Delta C)^2 \approx \overline{x^2} \cdot (\Delta k)^2$$

4. Conclusion and Evaluation

After the linearization the slope of the plot is $\frac{4\pi^2}{k} = 1.5751$, from which we can tell that the spring elasticity constant is $k = 25.1 \text{ N/m}$. On the other hand, the y-intercept is

$\frac{4\pi^2 \cdot \beta \cdot m_{spring}}{k} = 0.015$. After substituting the $m_{spring} = 0.027 \text{ g}$ and $k = 25.1 \frac{\text{N}}{\text{m}}$ in to the equation, we get the coefficient $\beta = 0.35$.

The error for the slope of the plot was calculated 0.037 N/m, hence an error of $\pm 0.61 \text{ N/m}$ in the constant k. The y-intercept has an error of 0.025, which implies that an error of ± 0.59 in β .

We get $k = 25.1 \pm 0.61 \text{ N/m}$

$\beta = 0.35 \pm 0.59$